

Inorganic Chemistry

including bioinorganic chemistry

Inorg. Chem., 1997, 36(11), 2301-2308, DOI:[10.1021/ic960838g](https://doi.org/10.1021/ic960838g)

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Table S1. Hydrogen Coordinates ($\times 10^4$) and Isotropic Displacement Parameters ($\text{\AA}^2 \times 10^3$) for **1**

	x	y	z	U(iso)
H(11A)	8324(15)	7510(7)	9284(2)	308
H(11B)	9269(15)	8332(7)	9018(2)	308
H(21A)	10571(14)	7338(10)	9424(4)	240
H(21B)	11352(14)	7986(10)	9139(4)	240
H(31A)	10264(23)	5193(14)	9161(4)	308
H(31B)	9304(23)	5593(14)	8941(4)	308
H(31C)	8516(23)	4942(14)	9228(4)	308
H(12A)	8994(25)	10130(22)	9678(2)	228
H(12B)	10886(25)	11268(22)	9672(2)	228
H(22A)	11321(33)	9395(29)	9862(2)	130
H(22B)	9495(33)	7992(29)	9813(2)	130
H(32A)	9739(41)	8765(25)	10264(3)	122
H(32B)	8609(41)	9337(25)	10122(3)	122
H(32C)	10462(41)	10620(25)	10183(3)	122

Table S2. Anisotropic displacement parameters ($\text{\AA}^2 \times 10^3$) for **1**
The anisotropic displacement factor exponent takes the form:
 $-2 \pi^2 [h^2 a^{*2} U_{11} + \dots + 2 h k a^* b^* U_{12}]$

	U11	U22	U33	U23	U13	U12
Mn	41(1)	41(1)	74(2)	0	0	21(1)
Cr	39(1)	39(1)	52(1)	0	0	20(1)
O(1)	50(2)	54(2)	76(4)	-11(3)	7(3)	29(2)
O(2)	46(2)	56(2)	73(4)	-21(3)	-10(2)	24(2)
O(3)	48(2)	48(2)	51(3)	-9(2)	5(2)	17(2)
O(4)	44(2)	44(2)	51(3)	-13(2)	-4(2)	17(2)
C(1)	47(3)	40(2)	43(3)	1(3)	8(3)	27(2)
C(2)	49(3)	35(3)	58(5)	-8(3)	-12(4)	21(2)
N(1)	82(4)	82(4)	40(5)	0	0	41(2)
C(11)	380(20)	344(20)	182(16)	-76(18)	-17(20)	283(17)
C(21)	240(18)	111(11)	208(25)	1(13)	3(19)	57(12)

C(31)	351(16)	281(16)	104(9)	18(13)	-12(13)	248(13)
C(32)	69(13)	52(14)	81(13)	-14(12)	-28(21)	-1(10)

Table S3. Hydrogen Coordinates ($\times 10^4$) and Isotropic Displacement Parameters ($\text{\AA}^2 \times 10^3$) for **2**

	x	y	z	U(iso)
H(11A)	5173(15)	1216(6)	6652(6)	344
H(11B)	6986(15)	1911(6)	6445(6)	344
H(11C)	6614(9)	1753(7)	6393(6)	222
H(11D)	8295(9)	2676(7)	6792(6)	222
H(21A)	5111(14)	-597(8)	7303(9)	407
H(21B)	6738(14)	586(8)	7720(9)	407
H(21C)	6037(15)	373(9)	7712(9)	258
H(21D)	7759(15)	647(9)	7494(9)	258
H(31A)	8008(16)	441(16)	6477(12)	247
H(31B)	7636(16)	-865(16)	7101(12)	247
H(31C)	4992(16)	-1839(14)	7026(14)	178
H(31D)	5598(16)	-704(14)	6318(14)	178
H(41A)	6601(33)	-2162(25)	5946(14)	432
H(41B)	5512(33)	-1353(25)	5890(14)	432
H(41C)	5159(33)	-2661(25)	6510(14)	432
H(41D)	6606(22)	-2529(27)	6279(27)	388
H(41E)	7476(22)	-1866(27)	7052(27)	388
H(41F)	8072(22)	-743(27)	6341(27)	388
H(12A)	6724(14)	2318(17)	8165(4)	356
H(12B)	7828(14)	4187(17)	8166(4)	356
H(22A)	4694(19)	2422(33)	8678(6)	134
H(22B)	5437(19)	4274(33)	8498(6)	134
H(32A)	7486(25)	4905(29)	9355(8)	142
H(32B)	5784(25)	4214(29)	9702(8)	142
H(42A)	7493(57)	3951(67)	10486(30)	745
H(42B)	7737(57)	2836(67)	9914(30)	745
H(42C)	6035(57)	2219(67)	10293(30)	745

Table S4. Anisotropic displacement parameters ($\text{\AA}^2 \times 10^3$) for **2**
The anisotropic displacement factor exponent takes the form:
 $-2 \pi^2 [h^2 a^{*2} U_{11} + \dots + 2 h k a^* b^* U_{12}]$

	U11	U22	U33	U23	U13	U12
Fe	56(1)	56(1)	68(1)	0	0	28(1)
Mn	52(1)	52(1)	80(1)	0	0	26(1)
O(1)	63(2)	59(2)	66(3)	-17(2)	-14(2)	27(2)
O(2)	66(2)	67(2)	80(4)	-18(3)	-9(3)	38(2)
O(3)	66(2)	62(2)	69(3)	-10(2)	-21(3)	35(2)
O(4)	75(3)	74(3)	72(3)	-15(3)	-19(3)	39(2)
C(1)	45(3)	61(3)	64(5)	4(3)	0(3)	27(2)
C(2)	52(3)	36(3)	67(5)	-9(3)	-8(3)	13(3)

Table S5. Crystal Data and Structure Determination Parameters ^a

	1	2
formula	C ₁₈ H ₂₈ CrMnNO ₁₂	C ₂₂ H ₃₆ FeMnNO ₁₂
fw	557.35	617.31
T, K	295	295
crystal system	rhombohedral	hexagonal
space group	R3c (No. 161)	P6 ₃ (No. 173)
a, Å	9.363(3)	9.428(2)
b, Å	9.363(3)	9.428(2)
c, Å	49.207(27)	17.827(8)
$\alpha, \beta, \gamma, ^\circ$	90, 90, 120	90, 90, 120
V, Å ³	3736(3)	1388.1(7)
Z	6	2
calcd density, g cm ⁻³	1.486	1.477
μ , cm ⁻¹	10.01	10.37
transm. fact. ^b range	0.6669-0.8293	0.7683-0.8404
F(000)	1728	644
crystal size, mm	0.44x0.43x0.15	0.37x0.28x0.20
2 θ range (deg)	2-72	2-60
scan speed (deg/min)	1.83-16.5	1.65-16.5
index range, h,k,l	-15/15, 0/15, -28/0	-13/13, 0/13, 0/25
reflens measured	6401 (incl. 60 Standards)	4442 (incl. 45 Standards)
unique reflens	2546	1395
R _{int}	0.137	0.087
reflens used	2503	1395
reflens with I>2 σ (I)	917	674
parameters	109 (19 restraints)	113 (36 restraints)
goodness-of-fit on F ²	1.101	1.149
final R ₁ ^c and wR ₂ ^d indices based on F and F ² , I>2 σ (I)	0.068, 0.162	0.071, 0.185
R ₁ and wR ₂ indices (all data)	0.214, 0.255	0.153, 0.245
weight multiplier n ^e	0.0819	0.0944
absolute struct. parameter ^f	0.04(10)	0.12(12)
2nd extinction ^g parameter	0.0000(1)	0.00(1)
max shift/esd	-0.49 (extinction param.) ^h	-1.32 (extinct. param.) ^h
min, max peaks (e/Å ³)	-0.62, 0.55	-0.45, 0.52

^a Data for both compounds were collected on an Enraf-Nonius CAD-4 diffractometer using graphite monochromatized Mo K α radiation ($\lambda = 0.71073$ Å). Unit cell parameters were obtained from the least-squares refinement of the θ angles of 25 reflections with $11.4^\circ < 2\theta < 24.0^\circ$ for **1** and $8.4^\circ < 2\theta < 30.2^\circ$ for **2**. ^b Absorption corrections based on seven (**1**) and 13 (**2**) indexed and measured crystal faces. ^c R_1 factor definition: $R_1 = \Sigma(|F_O| - |F_C|)/\Sigma|F_O|$. ^d SHELXL-93 wR_2 factor definition: $wR_2 = [\Sigma w(F_O^2 - F_C^2)^2/\Sigma w(F_O^2)^2]^{1/2}$. ^e Weighting scheme: $w = 1/[\sigma^2(F_O)^2 + (np)^2 + 0.00p]$, $p = (\max(F_O^2) + 2F_C^2)/3$. ^f Expected values for the absolute structure parameter are 0 (within 3 esd's) for correct and +1 for inverted absolute structure. ^g The secondary extinction parameter x was refined by least-squares, where F_O is multiplied by $-1/4k[1 + 0.001xF_C^2\lambda^3/\sin(2\theta)]$. ^h Relatively high max shift/esd values are due to a large number of weak reflections in **1** and **2** and the disordered cation in **2**. It may be noted that about 64% of the intensity data in **1** and about 52% of intensities in **2** are weak [$I < 2\sigma(I)$], but all data were used in the refinement with F^2 , thus R-values are relatively high.